CSE 400: Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe
Ahmedabad University

Q1. Random point on a line segment of length L

Statement: A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than 1/4.

Step 1: Interpretation

Let the line segment be represented by the interval:

$$(0, L)$$

Let the random point be denoted by the random variable:

$$X$$

Since the point is chosen at random on the line segment:

$$X \sim \text{Uniform}(0, L)$$

Thus, the probability density function is:

$$f_X(x) = \begin{cases} \frac{1}{L}, & 0 \leq x \leq L \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Define shorter and longer segments

Left segment:

$$X$$

Right segment:

$$L - X$$

Define:

$$S = \min(X, L - X)$$

$$G = \max(X, L - X)$$

Condition:

$$\frac{S}{G} < \frac{1}{4}$$

Step 3: Case 1

If:

$$X \leq \frac{L}{2}$$

Then:

$$S = X, \quad G = L - X$$

Condition:

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides:

$$4X < L - X$$

$$5X < L$$

$$X < \frac{L}{5}$$

Step 4: Case 2

If:

$$X > \frac{L}{2}$$

Then:

$$S = L - X, \quad G = X$$

Condition:

$$\frac{L - X}{X} < \frac{1}{4}$$

Multiply both sides:

$$4(L - X) < X$$

$$4L - 4X < X$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Step 5: Compute probability

$$P(X < L/5) = \frac{L/5}{L} = \frac{1}{5}$$

$$P(X > 4L/5) = \frac{L - 4L/5}{L} = \frac{1}{5}$$

Total probability:

$$\frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$\boxed{\frac{2}{5}}$$

Q2. Normal distributions in white and black image regions

White region:

$$\mu = 4, \quad \sigma^2 = 4$$

Black region:

$$\mu = 6, \quad \sigma^2 = 9$$

Fraction black:

$$\alpha$$

Fraction white:

$$1 - \alpha$$

Observation:

$$X = 5$$

Step 1: Density functions

White density:

$$f_W(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x-4)^2}{8}\right)$$

Black density:

$$f_B(x) = \frac{1}{\sqrt{18\pi}} \exp\left(-\frac{(x-6)^2}{18}\right)$$

Step 2: Equal error condition

$$(1-\alpha) f_W(5) = \alpha f_B(5)$$

Substitute:

$$(1-\alpha)\frac{1}{\sqrt{8\pi}} \exp(-1/8) = \alpha \frac{1}{\sqrt{18\pi}} \exp(-1/18)$$

Step 3: Solve

$$\frac{1-\alpha}{\alpha} = \frac{\sqrt{8\pi}}{\sqrt{18\pi}} \exp(1/8 - 1/18)$$

Cancel π:

$$= \frac{\sqrt{8}}{\sqrt{18}} \exp\left(\frac{5}{72}\right)$$

$$= \frac{2}{3} \exp\left(\frac{5}{72}\right)$$

Thus:

$$\boxed{\alpha = \frac{1}{1 + \frac{2}{3}e^{5/72}}}$$

Q3(a). Fire station location on finite road

$$X \sim \text{Uniform}(0, A)$$

Minimize:

$$E[|X - a|]$$

Step 1: Expectation

$$E[|X - a|] = \int_0^A |x - a| \frac{1}{A} dx$$

Split integral:

$$= \frac{1}{A} \left[\int_0^a (a - x) dx + \int_a^A (x - a) dx \right]$$

First integral:

$$= \frac{a^2}{2}$$

Second integral:

$$= \frac{A^2}{2} - aA + \frac{a^2}{2}$$

Total:

$$E = \frac{1}{A} \left[a^2 - aA + \frac{A^2}{2} \right]$$

Step 2: Differentiate

$$\frac{dE}{da} = \frac{1}{A}(2a - A)$$

Set equal to zero:

$$2a - A = 0$$

$$a = \frac{A}{2}$$

$$\boxed{A/2}$$

Q3(b). Infinite road exponential distribution

$$f(x) = \lambda e^{-\lambda x}$$

Minimize:

$$E[|X - a|]$$

Split:

$$\int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Condition:

$$P(X \leq a) = P(X \geq a)$$

$$1 - e^{-\lambda a} = e^{-\lambda a}$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

$$\boxed{\frac{\ln 2}{\lambda}}$$